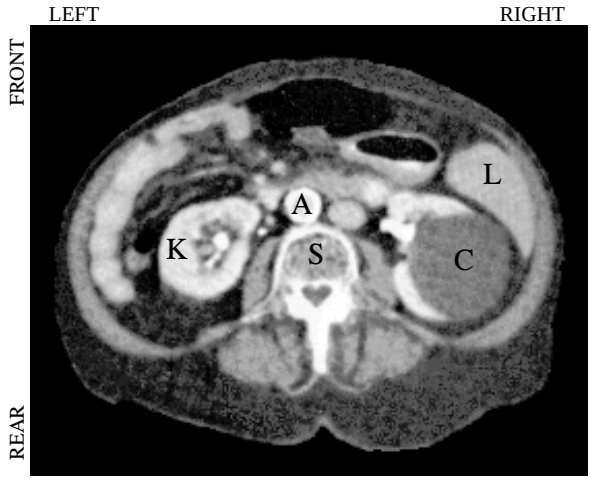

FIGURE 25-13
Computed tomography image. This CT slice is of a human abdomen, at the level of the navel. Many organs are visible, such as the (L) Liver, (K) Kidney, (A) Aorta, (S) Spine, and (C) Cyst covering the right kidney. CT can visualize internal anatomy far better than conventional medical x-rays.

along the beam's path. Materials such as bone and teeth block more of the x-rays, resulting in a lower signal compared to soft tissue and fat. As shown in the illustration, the source and detector assemblies are translated to acquire a **view** (CT jargon) at this particular angle. While this figure shows only a single view being acquired, a complete CT scan requires 300 to 1000 views taken at rotational increments of about 0.3° to 1.0°. This is accomplished by mounting the x-ray source and detector on a rotating gantry that surrounds the patient. A key feature of CT data acquisition is that x-rays pass *only* through the slice of the body being examined. This is unlike classical tomography where x-rays are passing through structures that you try to suppress in the final image. Computed tomography doesn't allow information from irrelevant locations to even enter the acquired data.

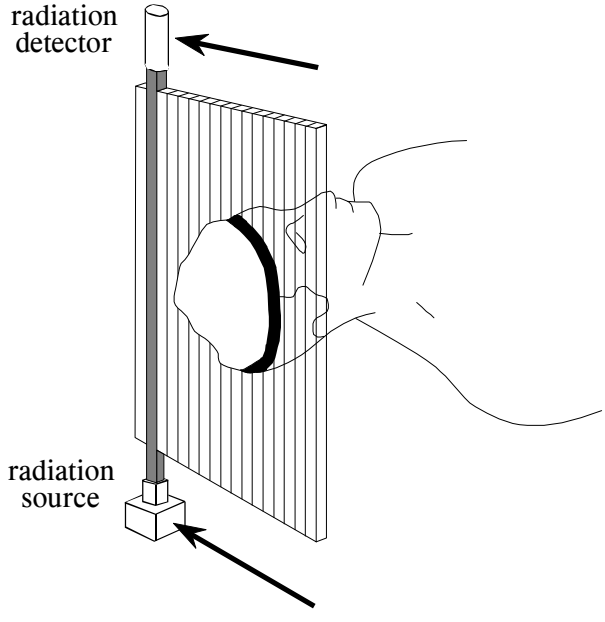

FIGURE 25-14
CT data acquisition. A simple CT system passes a narrow beam of x-rays through the body from source to detector. The source and detector are then translated to obtain a complete view. The remaining views are obtained by rotating the source and detector in about 1° increments, and repeating the translation process.